INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	OH facility OH_way_1411056682 -- station KOSU, America/New_York
Building	OSM 1411056682 OSM way 1411056682
Facade gap	not applicable -- one building, no second facade
Weather record	43,575 real hourly records from KOSU
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2024-02-28 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 16 of 24 hours with 1 mode change. That is 10 more than the reactive incumbent, at 0 unsafe hours against its 2. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	16 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	6 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s) 6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	16.110	18.0	16.110	0.842
01	mechanical	yes	switch budget	16.640	18.0	16.670	1.001
02	mechanical	yes	switch budget	16.665	18.0	16.670	0.952
03	mechanical	yes	switch budget	17.775	18.0	18.890	0.879
04	mechanical	no	dry-bulb	18.335	18.0	19.440	0.808
05	mechanical	yes	-	16.115	18.0	15.000	0.741
06	mechanical	yes	-	17.225	18.0	15.000	0.633
07	mechanical	no	dry-bulb	18.610	18.0	15.560	1.019

08	FREE	yes	-	16.665	18.0	13.330	1.564
09	FREE	yes	-	15.280	18.0	7.780	2.131
10	FREE	yes	-	15.275	18.0	6.110	2.303
11	FREE	yes	-	12.770	18.0	4.440	1.905
12	FREE	yes	-	8.335	18.0	2.780	1.420
13	FREE	yes	-	6.945	18.0	2.780	1.232
14	FREE	yes	-	5.275	18.0	2.220	1.022
15	FREE	yes	-	3.615	18.0	1.670	0.837
16	FREE	yes	-	3.055	18.0	1.110	0.997
17	FREE	yes	-	1.665	18.0	0.000	0.909
18	FREE	yes	-	0.555	18.0	-1.110	1.068
19	FREE	yes	-	-0.280	18.0	-1.670	1.159
20	FREE	yes	-	-1.110	18.0	-2.220	1.381
21	FREE	yes	-	-2.220	18.0	-3.330	1.361
22	FREE	yes	-	-2.500	18.0	-3.330	1.185
23	FREE	yes	-	-3.055	18.0	-3.890	0.971

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.110 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.640 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.665 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.775 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.335 C against a 18.0 C limit, so it fails by 0.335 C. It would take a limit 0.335 C higher, or a bound 0.335 C tighter, to change this hour.

05:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.115 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.225 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.610 C against a 18.0 C limit, so it fails by 0.610 C. It would take a limit 0.610 C higher, or a bound 0.610 C tighter, to change this hour.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.665 C, which is 1.335 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.564 C, and the margin is measured, not chosen: 1.564 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.280 C, which is 2.720 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.131 C, and the margin is measured, not chosen: 2.131 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.275 C, which is 2.725 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.303 C, and the margin is measured, not chosen: 2.303 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.770 C, which is 5.230 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.905 C, and the margin is measured, not chosen: 1.905 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.335 C, which is 9.665 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.420 C, and the margin is measured, not chosen: 1.420 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.945 C, which is 11.055 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.232 C, and the margin is measured, not chosen: 1.232 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.275 C, which is 12.725 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.022 C, and the margin is measured, not chosen: 1.022 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 3.615 C, which is 14.385 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.837 C, and the margin is measured, not chosen: 0.837 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 3.055 C, which is 14.945 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.997 C, and the margin is measured, not chosen: 0.997 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.665 C, which is 16.335 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.909 C, and the margin is measured, not chosen: 0.909 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.555 C, which is 17.445 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.068 C, and the margin is measured, not chosen: 1.068 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.280 C, which is 18.280 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.159 C, and the margin is measured, not chosen: 1.159 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -1.110 C, which is 19.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.381 C, and the margin is measured, not chosen: 1.381 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -2.220 C, which is 20.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.361 C, and the margin is measured, not chosen: 1.361 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -2.500 C, which is 20.500 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.185 C, and the margin is measured, not chosen: 1.185 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -3.055 C, which is 21.055 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.971 C, and the margin is measured, not chosen: 0.971 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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